

TL;DR: RAISE is **training-free** and **plug-and-play**, it refines prompt-image **alignment** using detailed **requirement engineering** and **verification** over **multiple rounds**, **adaptively scaling test-time compute** for T2I generation.

Background

Prompt-image alignment remains challenging

- Complex prompts often require **correct counts**, **attributes**, **relations**, **text rendering**, and **implicit intent**
- Existing scaling methods use **fixed budgets (non-adaptive)** or **sequential refinement**, limiting multi-round improvement
- Reflection-tuned methods require **extra training**, **curated special data**, and **model-specific tuning**

Goal: **improve T2I alignment** at test time, without training/fine-tuning or model modification. Achieve **adaptive scaling of test-time compute**

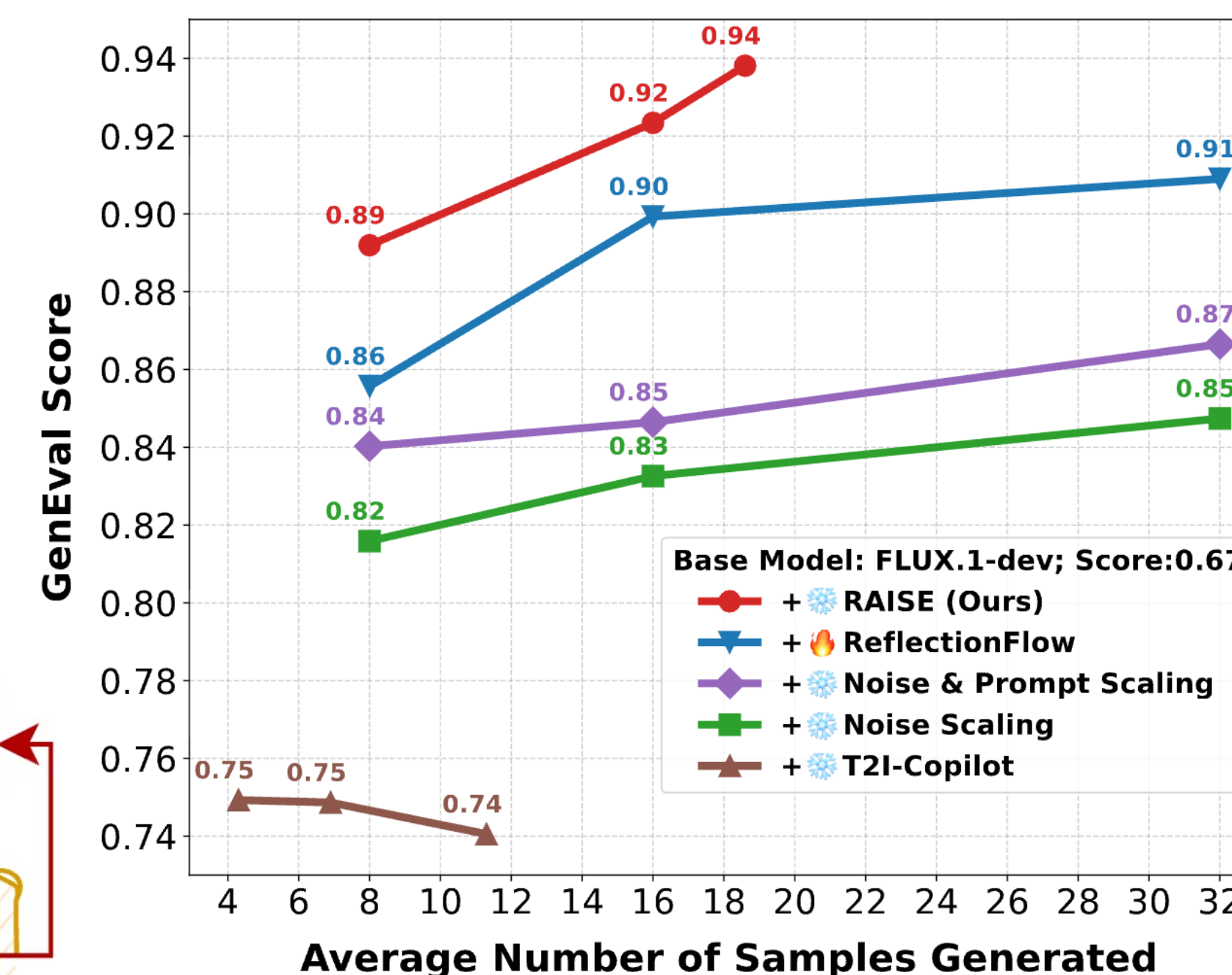
Methodology

RAISE operates as a **multi-agent system**

- Analyzer Agent:** extract **explicit** and **implicit requirements** from prompt into a **checklist**
- Rewriter Agent:** generate improved text **prompts** and targeted image **editing instructions**
- Evolutionary Refinement:** create diverse candidates via **resampling**, **rewriting**, and **editing**
- Verifier Agent:** check candidates by answering **binary requirement questions** and **visual evidence** extracted by tools
- Requirement-Adaptive Scaling:** continue to the next round only when **major requirements** remain unsatisfied

Results

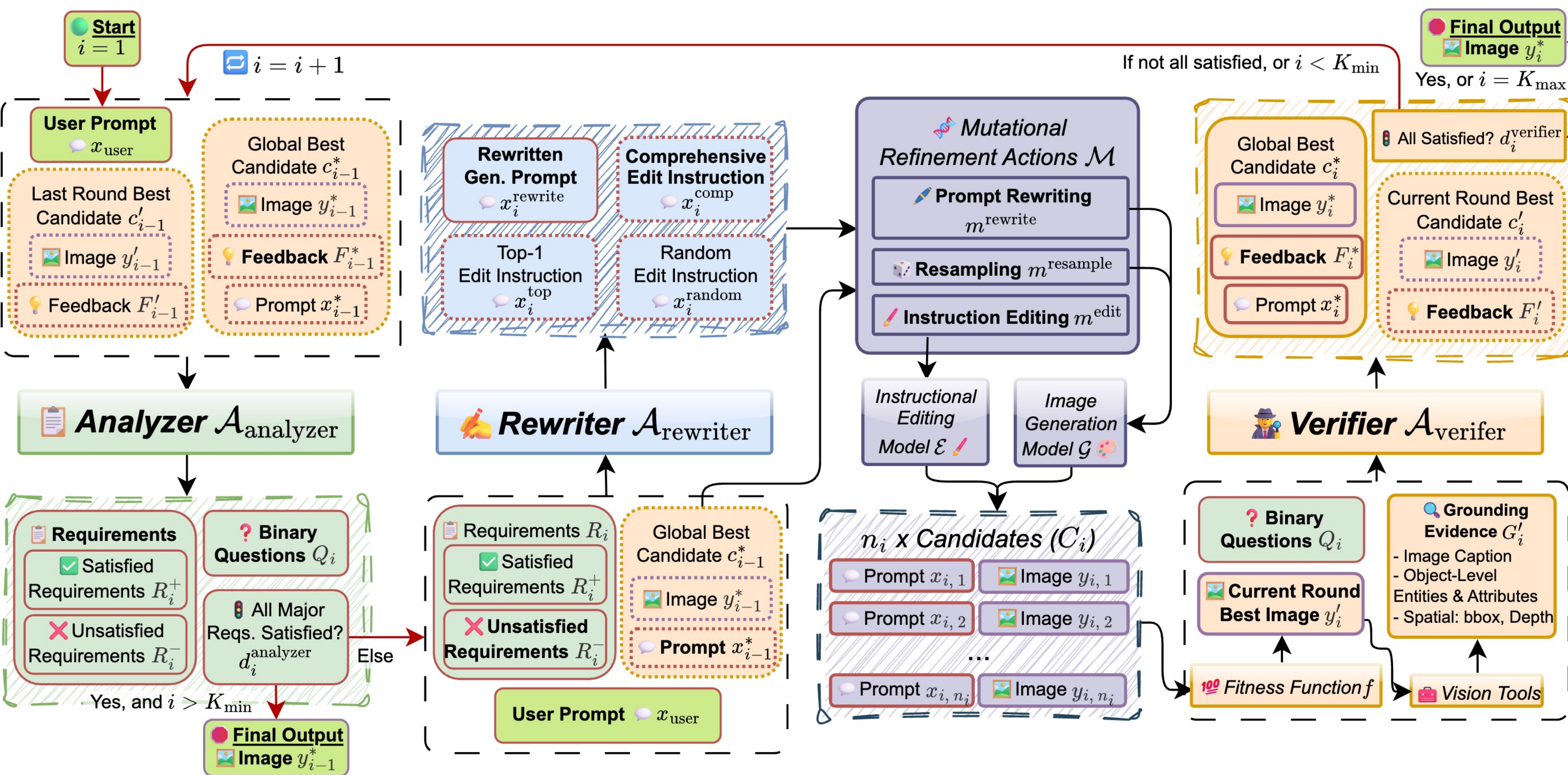
- Achieve State-of-the-art T2I alignment:** RAISE achieves 0.94 GenEval and 0.885 DrawBench VQAScore, beating training-based and training-free baselines and even outperforms UMMs (Qwen-Image-RL, BAGEL, and GPT-Image-1) on GenEval
- More efficient:** **reduces generated samples by 30-40%** and **VLM calls by 80%** vs. training-based reflection-tuned baselines



Our Method - RAISE (with FLUX.1-dev)

Prompt: "McDonald's Church"

Round	Analyzer	Rewriter	Tool-Augmented Verifier
Round 1	Requirements - Church main subject - Church elements: bench, stained glass - People - Logo (golden arch) - Text signage - ... color, style ...		✓ Church ✓ Logo ✓ People ... Text is not "McDonald's Church" ... environment does not clearly indicate it is a place of worship
Round 2	✓ ... ✗ Text should say 'McDonalds Church' to reinforce theme	- Specify 'McDonalds Church' should be prominently displayed - Specify font & size	✓ ... ✗ Text is not "McDonald's Church" ✗ environment does not clearly indicate it is a place of worship
Round 3	✓ ... ✗ 'Mcrolal's Hurch' is incorrect ✗ place of worship with people inside/outside	Replace 'Mcrolal's Hurch' with 'McDonalds Church'... add people ...	✓ ... ✗ main subject is a McDonald's restaurant, not an actual church ✗ place of worship
Round 4	✗ Not clearly a place of worship, improve by adding people ✗ Text prominent	Generation Prompt: - Add details of people's activities (socializing) - Text in bold	✓ All Satisfied!



Comparison of results across different models and prompts:

Prompt	FLUX.1-dev	T2I-Copilot	ReflectionFlow	RAISE (Ours)
a photo of two wine glasses (Implicit Requirement: glasses of wine ✗, wine glasses ✓)				
Three cats and three dogs sitting on the grass. (Counting / Multiple Object)				
A brown bird and a blue bear. (Rare Attribute / Combination)				
a photo of a train right of a dining table (Position / Spatial Composition)				
New York Skyline with 'Text to Image' written with fireworks on the sky. (Generating Text)				